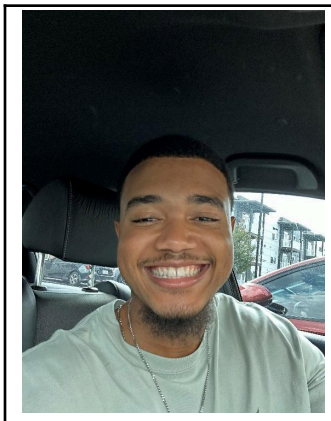
SP-108 – Red ASM

CS 4850 – Section 02 – Spring 2026

January 25th, 2026

Professor Sharon Perry

 Kimani Gordon Team Leader Developer	Adam Llado Documentation	Alec Sundby Documentation	Austin Abeln Developer

Project Team

Roles	Name	Role	Contact (Cell Phone)
Project owner	Sharon Perry	Owner/Facilitator	770-329-3895
Team leader	Kimani Gordon	Developer	912-248-5278
Team members	Adam Llado	Documentation	770-833-7620 allado@students.kennesaw.edu
	Alec Sundby	Documentation	770-815-2319
	Austin Abeln	Developer	770-731-3570
Advisor / Instructor	Sharon Perry	Facilitate project progress; advise on project planning and management.	770-329-3895

1.0 Project Overview / Abstract (Research)

Background:

Modern Enterprise networks are getting more and more fragmented due to rapid adoption of cloud services, third-party integration, and remote work-infrastructure. This leads to an unmonitored external

attack surface in which hackers can exploit. Traditional point-in-time vulnerability scans fail to account for "Shadow IT"—unmanaged assets like forgotten subdomains that provide low-resistance entry points for adversaries.

Objectives: The primary objective of this project is to develop an automated Attack Surface Management (ASM) framework that adopts an "outside-in" attacker's perspective to achieve three core goals:

Continuous Asset Discovery:
Automate the identification of both known and unknown assets (subdomains, IPs, and web services).

Risk Quantification:
Perform recursive vulnerability analysis on newly discovered assets to identify exploitable misconfigurations and high-severity CVEs.

Automated Remediation Pipeline:
Integrate discovery data with a security operations center (SOC) workflow to provide real-time alerting and incident response via a SOAR (Security Orchestration, Automation, Response) platform.

Scope: This project focuses on the External Attack Surface. The technical scope includes:

- Reconnaissance Engine: Utilizing *subfinder* for recursive DNS discovery and *httpx* for technology fingerprinting.
- Detection Layer: Implement *nuclei* for template based vulnerability scanning.
- Data Management: Developing a python based engine with a SQLite back-end to track and log data assets (subdomains, etc…) (new vs. existing).
- SOC integration: Configuring a cloud-hosted Wazuh (SIEM) manager to ingest JSON – formatted scan logs and Shuffle (SOAR) instance to automate Discord notifications and VirusTotal threat intelligence enrichment.

2.0 Project website

https://github.com/CS4850-ASMCyberProject/CS4850_Red_ASMCyberProject

Deliverables

Planning and Requirements (SDLC Phase: Plan/Define)

- Software Requirement Specification: A document defining the technical requirements for tool integration and data flow between the Python scanner and Wazuh.
- ASM Rules of Engagement: An ethical framework and scope document defining authorized target domains and rate limiting constraints.

Design and Architecture (SDLC Phase: Design)

- System Architecture Diagram: Visual representation of the data pipeline from local scanning nodes to the Oracle Cloud – hosted SOC brain.
- Database Schema: Documentation of the SQLite table structure used for asset state persistence.

Implementation & Testing (SDLC Phase: Implement/Test)

- Master ASM Python Script: Modular source code for automated discovery and scanning.
- Custom Wazuh Ruleset : XML based detection rules designed to identify “NEW_ASSET” and “VULN_FOUND” events.
- Shuffle SOAR Playbook: Exported workflow files for the VirusTotal-to-Discord automation loop.
- Documentation of Attack Simulation results against controlled, vulnerable targets.

Final Package (SDLC Phase: Maintain/Deploy)

- Final Technical Report: Comprehensive document in IEEE format summarizing objectives, synthesis of tools, and project evaluation.
- GitHub Organization Repository: A repository containing all commented source code, README documentation, and setup scripts.
- Public Project Website: A site containing the project summary, GitHub links, and the final presentation video.
- Final Presentation Video: A technical walkthrough demonstrating the end-to-end detection of a "Shadow IT" asset.

Group Meeting Schedule Date/Time

Jan. 30th, 2026

Collaboration and Communication Plan

SP-108 – Red ASM: Main Communication Channels:

- Group Text Chat (disseminate weekly objectives, roles, and a place for troubleshooting and discussion)

- KSU Email (share important docs or files)

Project Schedule and Task Planning

Gantt Chart Estimated Hours:

Total work hours	**391**

Other Plans – Like Risk Assessment (if applicable)

Risk Assessment: Limit Virtual Machine Access

- Two virtual machines will be deployed, ASM_Target (Vulnerability Monitoring) & ASM_Manager (Threat Detection & Response)
- Working with multiple virtual machines could invite unauthorized host connection, therefore, only ASM_Target will be accessible from the internet with ssh and remote desktop port access being limited to group member ips.
- ASM_Target will be used as a bastion to ssh and remote desktop into ASM_Manager, ensuring that public access to the project domain is strictly limited to group members.

Risk Assessment: Broken VM Configurations

- Oracle Cloud is the manager of the project VMs with boot volumes (VM images) saved weekly to ensure that if misconfigurations are made, backups can be restored.
- One Team member will manage and keep track of the VMs on Oracle Cloud

Risk Assessment: Data & Scope Integrity to Avoid Broad Generalizations & False Positives

- ASM scan tools, SIEM, & SOAR managers will run 24/7, but a broad scope & design could lead to uninformative conclusions and false positives.
- Group members will discuss and devise a targeted approach to scanning, ensuring that the right tools will be used, and custom scripts will be implemented strategically.
- Data will be scraped from ASM scan logs and SIEM and SOAR tools, offering a robust conclusion to software and configuration drift over time as well as how attackers and defenders react to oppositional posturing to achieve their goals.

Version Control Plan

SP-108 Red ASM: Git version control.

Our documentation, JSON scan logs, python scripts, and any other important files and deliverables will be uploaded to the project's dedicated GitHub repository:

https://github.com/CS4850-ASMCyberProject/CS4850_Red_ASMCyberProject.git